STAT 412 QZ07 — Question 2

Problem

A soft-drink machine dispenses amounts that are approximately normal with mean 190 milliliters and standard deviation 12 milliliters. A sample of 9 drinks is taken. If $181 < \bar{x} < 199$, the machine is considered satisfactory; otherwise, the owner concludes that $\mu \neq 190$. Find the type I error probability when $\mu = 190$ and the type II error probability when $\mu = 178$.

Solution

The standard error is

$$\sigma_{\bar{x}} = \frac{12}{\sqrt{9}} = 4.$$

For a type I error, assume $\mu = 190$ and find the probability of falling outside the acceptable interval:

$$\alpha = P(\bar{x} \leq 181) + P(\bar{x} \geq 199).$$

$$z_1 = \frac{181 - 190}{4} = -2.25, \qquad z_2 = \frac{199 - 190}{4} = 2.25.$$

$$\alpha = 2P(Z \leq -2.25) \approx 2(0.0122) = 0.0244.$$

For a type II error when $\mu = 178$, find the probability of staying inside the acceptable interval:

$$\beta = P(181 < \bar{x} < 199 \mid \mu = 178).$$

$$z_1 = \frac{181 - 178}{4} = 0.75, \qquad z_2 = \frac{199 - 178}{4} = 5.25.$$

$$\beta = P(0.75 < Z < 5.25) \approx 1 - 0.7734 = 0.2266.$$

Final Answer

$\boxed{\text{(a) Type I error probability} = 0.0244. \quad \text{(b) Type II error probability when } \mu = 178 \text{ is } 0.2266.}$